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Increasing efficiency of the homogenization process for production of chitin nanofibers for barrier film applications

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ABSTRACT

Chitin nanofibers (ChNFs) are of interest for barrier materials but are often extracted by high pressure homogenization (HPH) with high energy utilization. We studied the influence of deacetylation (DA) and pressure on HPH of shrimp chitin and the resulting solution cast films. Deacetylation to 72% DA resulted in improved ChNF suspension and film light transmission, strain at break, and tensile strength compared to chitin with DA of 89%. The oxygen permeability (OP) of the films was not affected by the modification and remained at low values of 1.9–2.4 cm^3 μm/m^2/day/kPa. We also show that deacetylation enables HPH intensity to be reduced during extraction of ChNFs from crab shell chitin (63% lower pressure and 73% reduction in number of passes), while achieving a low OP. Deacetylation pretreatment reduces HPH process intensity required to achieve oxygen barrier properties in ChNF films.

1. Introduction

High pressure homogenization (HPH) is an effective method for size-reduction of natural fibers, including producing nanofibers from polysaccharides like cellulose and chitin (Dufresne, 2013; Ono, Ogura, Kaku, Fujisawa, & Isogai, 2020). Chitin is an abundant biological material that occurs in the form of nanofibrils and microfibers in skeletal structures of fungi, diatoms, sponges, arachnids, crustaceans and insects (Kumar, 2000; Brunner et al., 2009; Kaya et al., 2017; Machałowski et al., 2019; Tsurkan et al., 2021). Within crustacean exoskeletons chitin occurs as microfibrils consisting of individual nanofibers of 2–5 nm diameter and 300 nm length embedded in a protein matrix (Ifuku & Saimoto, 2012). These can be extracted as nanofibers of 10–20 nm width and 1 to 4 μm in length (Ifuku & Saimoto, 2012; Wu, Zhang, Girouard, & Meredith, 2014). Chitin exists as a random block co-polymer of β-(1-4)-linked *N*-acetyl-glucosamine and N-glucosamine (Zhang & Rolandi, 2017). Structurally, the backbone is similar to cellulose, except with an acetamide group ($–NHCOCH_3$) or amine ($–NH_2$) at the C2 position. The existence of amine groups allows protonation under acidic conditions, making chitin a cationic polyelectrolyte. Chitin nanofibers and nanocrystals are of interest because of their ability to be fabricated into films with excellent gas barrier properties (Satam et al., 2018; Wu et al., 2014), high tensile strength and Young's modulus (Ifuku & Saimoto, 2012; Zeng, He, Li, & Wang, 2012), and transparency (Ifuku et al., 2013; Ifuku & Saimoto, 2012; Satam et al., 2018; Wu et al., 2014).

Homogenization processes use shear and impact forces to reduce size of particles in colloidal suspensions, often driven by pressure (Lim & Gong, 2018). In HPH a suspension at high pressure is forced through an orifice or valve resulting in pressure drop, turbulence, and cavitation that provide energy for size reduction of the particles (Chowdhury, Clarkson, & Youngblood, 2018; Levy, Okun, & Shpigelman, 2020). Smaller nozzle diameters lead to greater pressure drop and hence greater shear forces on the particles. However, smaller nozzle diameters also place an upper limit on the particle size that can pass through the homogenizer without clogging the nozzle. HPH can separate large hierarchical fibers (10–50 μm depending on source) into smaller constituent nanofibers with diameters <100 nm that have high fiber aspect ratio close to the original source fiber (Nechyporchuk, Belgacem, & Bras, 2016). HPH is associated with elevated energy usage (Nechyporchuk et al., 2016; Spence, Venditti, Rojas, Habibi, & Pawlak, 2011; Zimmermann, Bordeanu, & Strub, 2010) and chemical pretreatments have been explored to reduce energy intensity required to defibrillate materials (Dufresne, 2013; Zimmermann et al., 2010). Homogenization of cellulose suspensions, closely related to chitin, has been widely studied and reviewed (Dufresne, 2013; Moon, Martini, Nairn, Simonsen, & Youngblood, 2011; Nechyporchuk et al., 2016; Spence et al., 2011;

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Zimmermann et al., 2010). These suspensions are usually cycled through the homogenizer multiple times (Spence et al., 2011) and the number of passes and pressure affects the nanofiber sizes produced (Abdul Khalil, Bhat, & Ireana Yusra, 2012; Davoudpour et al., 2015; Moon et al., 2011; Spence et al., 2011). For cellulose nanofibers (CNF), homogenization resulted in fibers with high specific surface area and lowered water vapor transmission rate compared to methods like microfluidization (Spence et al., 2011). HPH of chitin differs from that of cellulose, as an acidic medium is used to protonate surface amine groups and promote colloidal stability (Goodrich & Winter, 2007; Paillet & Dufresne, 2001; Salaberria, Fernandes, Diaz, & Labidi, 2015; Wu et al., 2014). In the case of cellulose, chemical pretreatment methods like acid hydrolysis, enzyme-based pretreatment, and 2,2,6,6-tetramethylpiperidine-1-oxyl radical (TEMPO) mediated chemical oxidation have been effective in aiding homogenization (Dufresne, 2013; Paakko et al., 2007). Specifically, TEMPO-mediated oxidation introduces carboxylate groups that aid in suspension stability via electrostatic repulsion. Chemical oxidation of chitin was also used to introduce surface carboxylate groups that enhanced electrostatic repulsion, enabling preparation of chitin nanocrystals (ChNCs) by mechanical disintegration (Fan, Saito, & Isogai, 2008). These ChNCs are more crystalline than ChNFs and are have much smaller aspect ratios, with average width of 8 nm and lengths of around 300 nm for ChNCs (Fan et al., 2008) compared to 20 nm width and length of 1–4 μm for ChNFs (Wu et al., 2014). Another method of increasing surface charge is *N*-deacetylation of chitin to convert the acetamide groups to protonatable amino groups, which can be used to produce ChNFs (Fan, Saito, & Isogai, 2010). Complete *N*-deacetylation of chitin leads to chitosan, which is soluble in water under acidic conditions (Kumar, 2000). Conversely, acetylation of ChNFs has been used to improve compatibility with nonpolar solvents (Ifuku & Saimoto, 2012). Pretreatment procedures can have a significant impact on the energy usage during homogenization (Spence et al., 2011), but may affect properties of final films formed from the fibers, including optical transmission, tensile strength (Ifuku & Saimoto, 2012), and gas permeability (Spence et al., 2011). Qi et al. (Qi et al., 2013) improved homogenization nanofibrillation efficiency by altering the acid used for ChNF cationization, indicating the influence that acid type, dissociation equilibria, and pH have on fiber properties. Prior studies have also noted that pH varies during chitin homogenization (Wu et al., 2014). The pH varies with the number of protonated surface amine groups, which is a direct result of the new ChNF surface area formed during HPH. Previously, Fan et al. (2010) prepared chitin nano-whiskers by partial deacetylation of α-chitin fibrils followed by sonication, resulting in increased ChNF surface cationization.

This work reconfirmed the hypothesis of Fan et al. (2010) that *N*-deacetylation of chitin followed by HPH results in increased cationization of the resulting ChNFs. Secondly, the concept that optimized deacetylation allows a reduction in the number of HPH cycles required to produce fiber suspensions with equivalent or improved properties (*oxygen barrier, charge, light transmission and mechanical properties of formed films*) was tested and found to be true. To the authors' knowledge, this is the first study that demonstrates the interaction between chitin deacetylation pretreatment and HPH pressure and reports the resulting structure-property relationships for chitin films. We find that surface modification through deacetylation allowed significant reduction in the number of homogenization cycles (even at lowered HPH pressure) required to produce fiber suspensions with equivalent or better barrier and mechanical properties than un-optimized counterparts.

2. Experimental

2.1. Materials

Commercial chitin, purified from shrimp exoskeleton according to the manufacturer, was purchased from Alfa Aesar (Ward Hill, MA). ACS grade NaOH pellets were obtained from VWR (Radnor, PA). Crab shells were purchased from Neptune's Harvest (Gloucester, MA). HCl and NaOH were purchased from EMD Chemical Inc. (Burlington, MA). Glacial acetic acid was purchased from Sigma Aldrich (St. Louis, MO). Deionized (DI) water was obtained from a Thermo Scientific GenPure UV/UFxCAD-plus water purification system. Research-grade oxygen and ultra-high purity nitrogen were purchased from Airgas (Atlanta, GA).

2.2. Deacetylation of commercial chitin

Commercial shrimp-based chitin was deacetylated by an established procedure (Chang, Tsai, Lee, & Fu, 1997) Deacetylation of acetyl groups was kept below 50% to prevent production of chitosan. Deacetylation utilized a 35 wt% NaOH solution and a 25 ml/g chitin loading in this solution. The mixture was refluxed at 110 °C for 100 min (targeting 20% deacetylation) or 150 min (targeting 30% deacetylation). For the 0% deacetylation target, no deacetylation step was utilized.

2.3. Crab shell chitin purification and deacetylation

The authors' previous works have used crab shell chitin extensively. In order to provide continuity with this prior work, crab shell chitin was purified per previous studies (Satam et al., 2018; Satam et al., 2019; Wu et al., 2014) and deacetylated by using the same procedures utilized for commercial shrimp chitin. Further details can be found in the supplementary information section.

2.4. Extraction of ChNFs

Suspensions containing 0.5 wt% purified chitin in DI water from each preparation above were adjusted to pH 3.0 using glacial acetic acid. The suspension was homogenized in a Mini DeBEE Homogenizer (BEE International, South Easton, MA). Because of suspension heating due to shear in the nozzle, a water cooled heat exchanger integrated into the homogenizer (product cooler) was used to keep the suspension below 35 °C. Chitin suspensions with different DA values were homogenized using an established procedure (Wu et al., 2014). The HPH was carried out in two stages: 20 passes at 1034 bar with a 0.2 mm nozzle followed by 10 passes at 1516 bar utilizing a 0.13 mm nozzle.

To explore pressure effects, the 70% DA target commercial chitin was homogenized in a single sequence at constant pressure. Two different pressures were explored: a low pressure (LP) of 551 bar and a high pressure (HP) of 1723 bar. To test applicability of results from shrimp chitin to other sources, unmodified crab shell chitin was homogenized in two stages at 1034 bar and 1516 bar as a control. Deacetylated crab shell chitin with a targeted DA of 70% was homogenized at a reduced number of homogenization passes at a pressure of 551 bar, as suggested by the optimization on commercial shrimp chitin (discussed in results below).

2.5. ChNF and suspension characterization

Three samples were drawn at every pass during the homogenization process. The pH of the suspensions at each pass was obtained for three samples per pass using a Mettler Toledo Seven Excellence S400 pH meter. When standard deviation of the measurements was less than 0.01 log units, a standard error of 0.01 was assumed for the measurements corresponding to the least count of the instrument. Similarly, light transmission readings of suspensions at 550 nm for each pass was obtained in triplicate using a Shimadzu UV-1800 UV/Vis Spectrophotometer. The DA of the chitin batches was determined by potentiometric titration. A volume of 25 ml of homogenized commercial chitin suspension was ion exchanged using 300 ml (bed volume) of anion exchange resin. The resin was washed with 200 ml DI water to remove entrapped chitin. 20 ml of 0.25 N HCl was added to the resulting suspension, which was then titrated against 0.5 N NaOH using the same pH meter noted above. The inflection points determined from the pH vs.

volume of titrant curve were then used to determine the amount of NaOH consumed. The NaOH equivalents of HCl in the suspension was subtracted from the total consumed NaOH to yield the amount of NaOH consumed by the protonated chitin. The NaOH equivalents consumed divided by the mass of the starting material gives m [mol NH_2/g chitin (dry basis)], which was used to determine the DA by

$$\mathrm{DA}\ [\%] = 100 \times \frac{203.21}{42.04 + \frac{1}{m}} \quad (1)$$

where 203.21 [g/mol] is the molecular mass of chitin monomer and 42.04 [g/mol] is the difference in molecular mass between chitin and chitosan (deacetylated chitin) monomers. The titrations were repeated 3× for each sample. FTIR analysis of deacetylated chitin is also presented in Fig. S1 of supplementary information section.

Particle size distributions of ChNFs upon homogenization were characterized by AFM imaging by using a Bruker Icon AFM. To produce these samples, the final ChNF suspensions (30 passes), were diluted to 0.001 wt% with pH 3 acetic acid containing 0.025 vol% Triton X-100 surfactant. 20 μl of the diluted suspensions were drop cast onto an ozone-treated silicon wafer and allowed to dry at 60 °C for 1 d. The length and number of particles were determined by analysis of the AFM height images with ImageJ. The width of the nanoparticles was estimated using the AFM height value.

The stability of homogenized ChNF suspensions was characterized by zeta potential measurements. 40 μl of 0.5 wt% ChNF suspensions were diluted with 20 ml DI water. A Malvern Zetasizer Nano was used to measure the zeta potential and the hydrodynamic diameter of ChNFs in diluted suspensions.

2.6. Preparation of solution cast shrimp and crab shell ChNF films

50 ml of the 0.5 wt% homogenized chitin suspension extracted from the 30th pass (prepared above) were cast in a poly(tetrafluoroethylene) dish (8.5 cm × 13 cm) and air-dried for 14 d at a temperature of 21 °C and relative humidity (RH) of 45% to yield free-standing films.

2.7. Characterization of solution cast ChNF films

The thickness of cast films was measured with a micrometer at >10 points per film. Film light transmission was measured at 550 nm using a Shimadzu UV-1800 UV/Vis Spectrophotometer. Tensile properties of films were determined by using a high-throughput mechanical characterization (HTMECH) puncture instrument described previously (Sormana, Chattopadhyay, & Meredith, 2005). Student's *t*-test was performed in MS Excel on the mechanical property datasets. 3 different spots were chosen on 3 film samples of size 1.5 cm × 1.5 cm. In some cases, the samples cracked and data points where neighboring cracks caused puncture test to fail were excluded from the analysis.

The O_2 permeability values of the films were obtained by using a MOCON OXTRAN 1/50 instrument at 50% RH. Films were humidified to soften and facilitate loading in the MOCON by using an Electrotech humidifier (Glenside, PA). Films were cut into ~9 cm^2 squares and covered with a mask that allowed measurement of 5 cm^2 area. The interval between measurements was 1 h with the final measurement taken at steady state. The criteria for steady state was an O_2 transport rate change of <1% or <0.05 $cm^3/m^2/d$, generally achieved in ~24 h. AFM (Bruker Icon) was utilized to characterize the top surface of ChNF films drop cast onto a silicon wafer and dried for 1 d. The size of structures were measured by visual analysis of the AFM phase images. Film root mean square roughness was determined by using Bruker Nanoscope Analysis software. The films were fractured and the cross section was imaged using a Zeiss Ultra 60 FE-SEM.

3. Results & discussion

3.1. Influence on zeta potential

Table 1 shows DA and Zetasizer measurements for chitin suspensions with different DA values after HPH. Untreated commercial chitin has a DA of about 89%. For the 80 and 70% target DA samples we obtained actual DA values of 79% and 72%, respectively. The zeta potential increases with decreasing DA, with 89% DA suspensions having zeta potential of 32 mV and 72% DA having zeta potential of 86 mV. An increase in surface charge of the ChNFs, indicated by increased zeta potential, is expected to increase suspension stability. Suspensions with zeta potential above 30 mV, such as the 79% and 72% DA samples, are usually considered to be stable (Kumar & Dixit, 2017; Lu & Gao, 2010). Table 1 also shows the hydrodynamic radius as determined by the Stoke-Einstein equation (Bauer & Schnapp, 2007). While this is an approximation to the size of rod-like fibers, it indicates that decreasing DA led to decreased particle size.

The length and width distribution of ChNF fibers from AFM for different DA values and homogenization pressures are presented in Fig. S2 of the supplementary section. Median values for different DA and pressures are summarized in Table 2.

A two-sample Kolmogorov-Smirnov (KS) Test was performed on the distribution lengths and widths and can be found in Tables S1 and S2 of the supplementary information. The distributions show that the deacetylation of chitin from 89% to 79% does not change the length distribution significantly. This trend is supported by examination of median values in Table 2, which range from 887 to 895 nm; however, further deacetylation to 72% reduces the length drastically to the 300–500 nm range as indicated by the KS test in Table S1, and by a median value of 428 nm in Table 2. This corresponds well to the hydrodynamic diameter in Table 1 where the sizes for the 89% and 79% DA chitin are similar while the size of the 86% DA chitin is the lowest. As indicated from the KS test in Table S2, increasing deacetylation leads to lower width particles indicating increased fibrillation of fibers for deacetylated (72 & 79% DA) samples as compared to the unmodified chitin (89% DA). For studying the influence of pressure, chitin with a 72% DA was homogenized at two different pressures: LP = 551 bar and HP = 1723 bar. The zeta potential of the suspensions was measured to be 60.0 ± 5.4 for LP and 86.1 ± 6.3 for HP suspensions, indicating "good" to "excellent "stability (Kumar & Dixit, 2017). The Zetasizer-determined size of the suspensions was measured to be 512 ± 76 nm for LP and 506 ± 90 nm for HP suspensions. Fig. S3 in the supplementary information shows the particle length and width distributions obtained from AFM analysis, and Tables S3 and S4 show the corresponding KS tests for significance, at LP and HP. The length and width distributions are statistically similar for both pressures as seen in Table 2. Thus, homogenization pressure does not liberate significantly shorter particles for HP versus LP samples, when a 72% DA chitin is used. While an increase in pressure to HP conditions generally is expected to reduce particle lengths, however the effective stabilization of particles due to the low DA is expected to enable higher homogenization efficiencies even at LP conditions, such that similar widths can be obtained at a lower pressure.

Table 1
Degree of acetylation, zeta potential and hydrodynamic diameter for ChNF samples produced from commercial chitin.

Actual DA (%)	Zeta potential (mV)	Hydrodynamic diameter (nm)
89.2 ± 3.2	31.7 ± 3.9	1680 ± 310
79.0 ± 3.6	54.5 ± 4.1	1410 ± 360
71.6 ± 3.4	85.7 ± 6.0	930 ± 250

Table 2
Median ChNF length and width from AFM showing effect of DA and homogenization pressure for single and two-stage homogenization.

Pretreatment/homogenization condition	Length (nm)	Width (nm)
89.2 ± 3.2% DA, two stage	895 ± 551	5.0 ± 2.3
79.0 ± 3.6% DA, two stage	887 ± 644	3.7 ± 2.5
71.6 ± 3.4% DA, two stage	428 ± 270	2.9 ± 1.0
71.6 ± 3.4% DA, single stage, LP (551 bar)	585 ± 199	4.0 ± 1.5
71.6 ± 3.4% DA, single stage, HP (1723 bar)	466 ± 185	3.8 ± 1.7

3.2. Influence of DA and pressure on HPH characterized by suspension pH

Fig. 1(a) shows the change in pH with number of homogenizer passes for commercial chitin samples with different degrees of acetylation. The biggest difference is seen with chitin having 89% DA and that with 79% DA. The final pH for 89% DA chitin is around 3.10 while that for 79% DA and 72% DA is 3.19 and 3.22 respectively. The pH change with homogenization mostly occurs between 0 and 15 passes with the change in pH being very gradual after 15 passes.

As homogenization proceeds the pH is expected to increase as new surface area is exposed. The newly exposed surfaces have amino groups that on contact with the acidic suspension medium get protonated from NH_2 to NH_3^+. Each NH_2 consumes one proton from the suspension, increasing its pH. The magnitude of pH change depends on the (i) surface area of the fibers in suspension and (ii) the average number of amino groups per gram of chitin, which is represented by its DA. Higher DA values correspond to lower number of amino groups and consequently the maximum pH increase is less for higher DA versus the chitin having lower DA. However, it must also be noted that such suspensions are charge stabilized. Having lower DA values is advantageous as higher cationization of fiber surfaces increases the repulsive electrostatic barrier to fiber aggregation. Consequently, in HPH, as fibrillation proceeds, the fibers tend to remain in suspension longer for chitin with lower DA thus increasing the efficiency of homogenization.

Fig. 1(b) shows the change in pH with number of passes for chitin samples processed at different homogenization pressures. Chitin processed at LP has a final pH of 3.33 which is lower than chitin processed with HP which has final pH of 3.42. With increasing pressure, the final pH is seen to increase, albeit to a lower extent when compared to pH changes with DA seen in Fig. 1(a). This is to be expected as the final distribution of particle sizes and widths (Fig. S3) are more similar to each other at LP versus HP, than for the samples in Fig. S2 at different DA. Since the starting DA of the samples is the same, the surface NH_2 groups per unit area are expected to be similar and the uptake of H^+ from the surrounding solution will be similar leading to similar end pH values.

3.3. Influence of DA and homogenization pressure on light transmission of suspensions and cast ChNF films

Fig. 2(b) shows the light transmission at 550 nm for suspensions of chitin nanofibers sampled during homogenization. The light transmission at pass (cycle) 0 is that of the chitin slurry after the deacetylation step. As the HPH process proceeds to higher numbers of cycles, light transmission decreases for the 89% DA (unmodified chitin) sample while it gradually increases for the 72% and 79% DA (deacetylated) samples. The light transmission eventually increases after 10 passes for the 89% DA sample but after 30 passes it reaches only 36%, whereas the 79% and 72% DA samples reach light transmission values of 74% and 81%, respectively.

Fig. 2(b) shows the light transmission at 550 nm for suspensions of chitin nanofibers sampled during the homogenization at LP and HP. The light transmission at pass 0 corresponds to an unhomogenized chitin slurry. As the process proceeds the light transmission increases for all samples. The light transmission of LP and HP samples approaches 87% and 84% after 5 passes, respectively. Compared to the dependence of light transmission on DA (Fig. 1(b)), the LP and HP samples produced on a sample at 72% DA, are similar. The similarity of light transmission and particle size distributions indicated by Tables 2, S3 and S4 suggests charge stabilization as a controlling influence in HPH of the chitin suspension, regardless of the pressure utilized for HPH.

The higher light transmission of lower DA chitin samples (89% and 79% DA) in Fig. 1(b) points towards more fibrillated ChNFs as indicated by smaller fiber widths, seen in Table 2. These observations are consistent with higher zeta potential driven by higher surface charge (Table 1) for the lower DA chitin samples.

Fig. 2(c) shows the light transmission at 550 nm of films cast from suspensions produced by homogenizing chitin with different DA values for 30 passes. The average light transmission value of 89% DA chitin films is the lowest at 19% and light transmission increases with decrease in DA values with 79% and 72% DA films having transmission values of

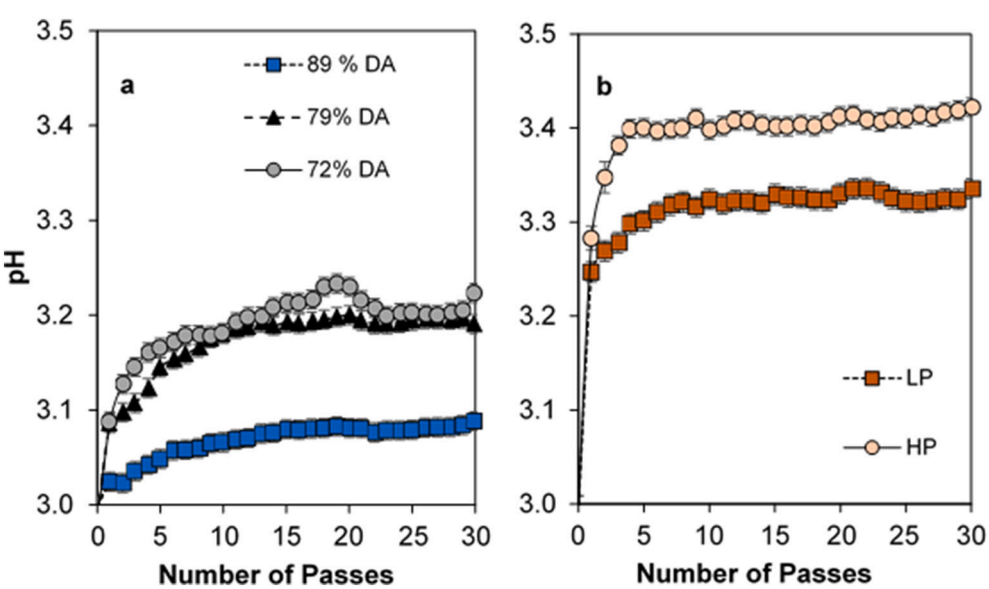

Fig. 1. Variation of suspension pH as a function of number of passes. The effect of changing DA for a two-stage homogenization is shown in (a) and of changing homogenization pressure for single-stage HPH of the 72 DA % sample is shown in (b) LP = 551 bar, HP = 1723 bar.

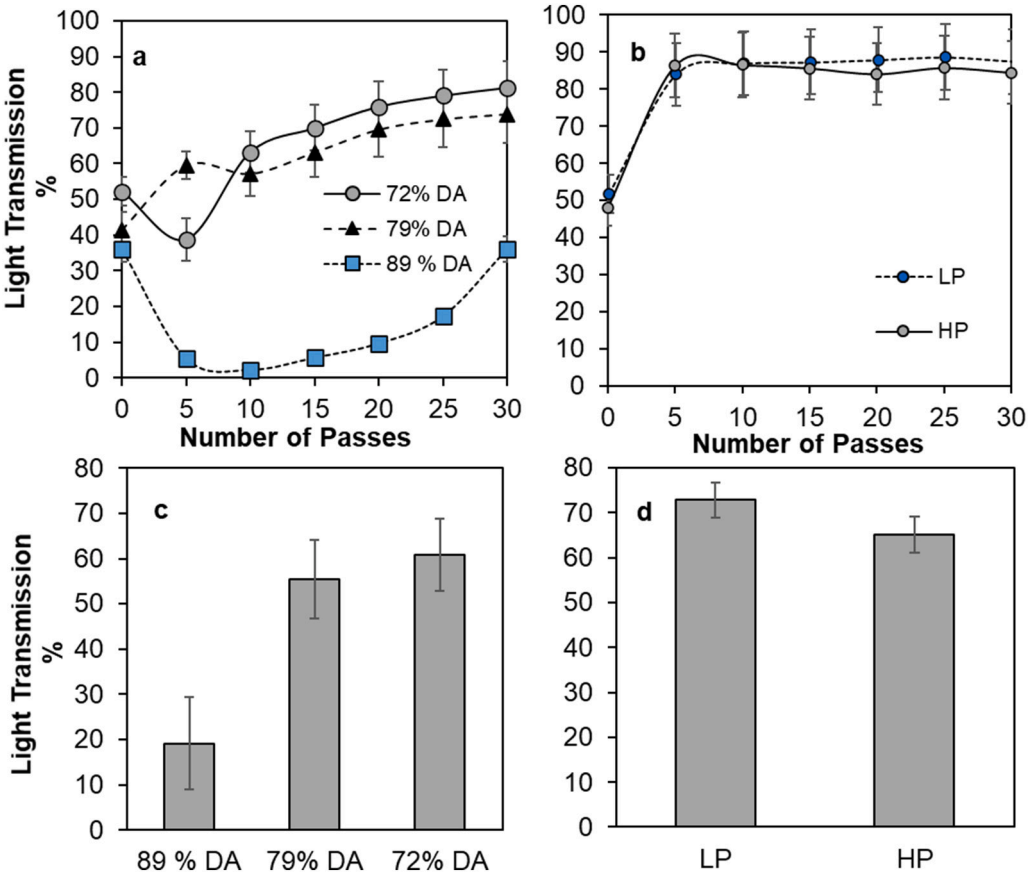

Fig. 2. Variation of suspension ((a), (b)) and film ((b), (d)) light transmission prepared from commercial ChNF samples. The effect of changing DA for a two-stage HPH is shown in (a) and (c) and changing homogenization pressure for the 72% DA sample is shown in (b) and (d). Permeability is measured at 50% RH and 23 °C. LP = 551 bar, HP = 1723 bar.

55% and 61%, respectively. Due to scattering effects, the intensity of transmitted light is roughly inversely correlated with particle size with the exact dependence on size being a function of factors like particle shape and beam geometry (Lothian & Chappel, 1951). For nonspherical particles like ChNFs, light transmission is a function of length and width. The increase in the transmission for homogenized suspensions with decrease in DA, seen in Fig. 2(c), points to reduced scattering and a shift in particle population towards lower width, length, or both. This is supported both by Fig. S2 and Tables S1 and S2 where particle width decreases with decreasing DA and particle length decreases significantly between 79% DA and 72% DA. The increase in light transmission of films is also related to the increase in suspension zeta potential with decreasing DA, which enables the particles to pack more efficiently during the drying process.

Fig. 2(d) shows the light transmission at 550 nm of films cast from suspensions produced by HPH at low and high pressure. The average light transmission value of LP chitin is 73 ± 4%. HP chitin samples have a similar light transmission value at 65 ± 4%. There is a minor decrease in transmission with increase in HP, possibly due to enhanced stabilization at the higher charge produced at HP (discussed above). This difference in light transmission is not statistically significant (p-value = 0.47).

3.4. Influence of DA and homogenization pressure on oxygen permeability of cast ChNF films

The thickness of 89% DA film was 84 ± 27 μm, while that for 79% DA was 66 ± 13 μm. The thickness of 72% DA film was 55 ± 15 μm. The reduction in measured thickness of films formed from fibers with increasing number of surface charges (lower DA) has been reported for TEMPO oxidized cellulose nanofibers (Rodionova, Eriksen, & Gregersen, 2012). The decrease in thicknesses with increasing cationization of the fibers can be attributed to electrostatic repulsion that promotes entropically driven packing, leading to lower film thicknesses for lower DA ChNFs.

Fig. 3(a) shows the oxygen permeability (OP) of films cast from suspensions produced by homogenizing chitin with different DA values for 30 passes. The OP values of the 89% DA, 79% DA and 72% DA samples are not significantly different considering the error bars and have an average of 2.1 ± 0.3 cm^3-μm/m^2/day/kPa. The opposing and coupled effects of changing length and charge, combined with a large variance in the particle width and length distribution, likely leads to this lack of detectable dependence of OP with DA%. The average OP values are, however, quite low for chitin with both high and low DA% and the result is similar to our previously reported results for crab shell-derived ChNFs with low degree of acetylation (Satam et al., 2019). In fact, the OP values reported here are about 5× lower than common packaging barrier materials like oriented poly(ethylene terephthalate) (Lange & Wyser, 2003), with OP ranging from 9 to 24 μm/m^2/day/kPa, and only about 1.5× higher similar in magnitude than high barrier poly(vinylidene chloride), with OP near 0.5 μm/m^2/day/kPa at the same conditions of temperature and humidity used here.

Fig. 3(b) shows the OP of films cast from suspensions produced by HPH at low and high pressure. The OP values of the HP samples are similar (1.6 ± 0.4 cm^3-μm/m^2/day/kPa) to the LP chitin samples with OP values of 1.4 ± 0.3 cm^3-μm/m^2/day/kPa. The particle length distributions encountered for different pressures per the KS test (Table S3) are similar and thus the aspect ratios of particles are largely unchanged

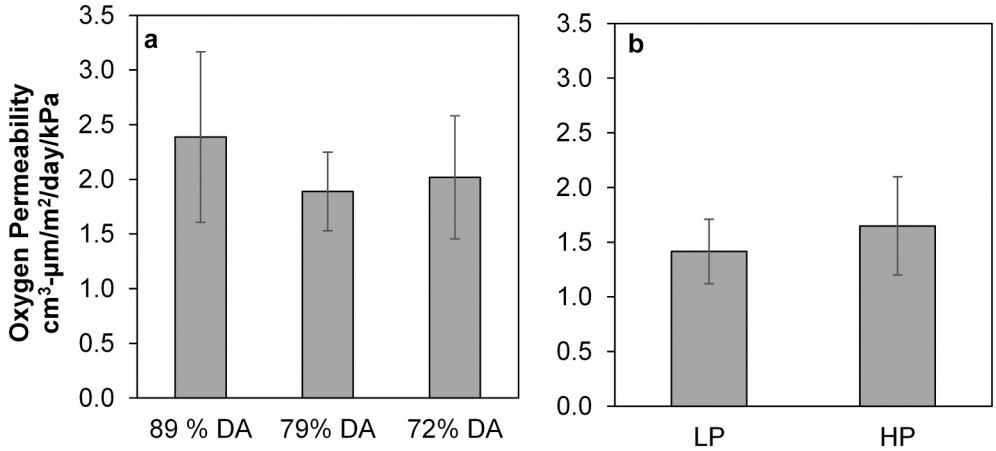

Fig. 3. Variation of oxygen permeability for dried cast films prepared from commercial shrimp-derived ChNF samples. The effect of changing DA for a two-stage HPH is shown in (a) and changing homogenization pressure for the 72% DA sample is shown in (b). Permeability is measured at 50% RH and 23 °C. LP = 551 bar, HP = 1723 bar.

with processing conditions. The OP values are thus expected to be similar due to similar charge distributions which is expected to produce similar alignment.

3.5. Influence of DA and pressure on cast film mechanical properties

Fig. 4(a) shows box plots of the ultimate tensile strength (UTS, the maximum value in stress versus strain) of films cast from ChNF suspensions produced by a two-stage HPH of commercial shrimp chitin with different DA values for 30 passes. The box-plot shows the median value at the intersection of the light and shaded areas, where the top of the light area is the 75th percentile and the bottom of the dark area is the 25th percentile. Error bars represent the total range of the data set. Tables S5 and S6 indicate two-tailed P values. No statistically significant differences in UTS were observed when comparing 89% and 79% DA, however higher values were observed with 72% DA samples (p-value $<$ 0.002 in both cases).

Fig. 4(b) shows the strain at break values of films. Similar to the UTS,

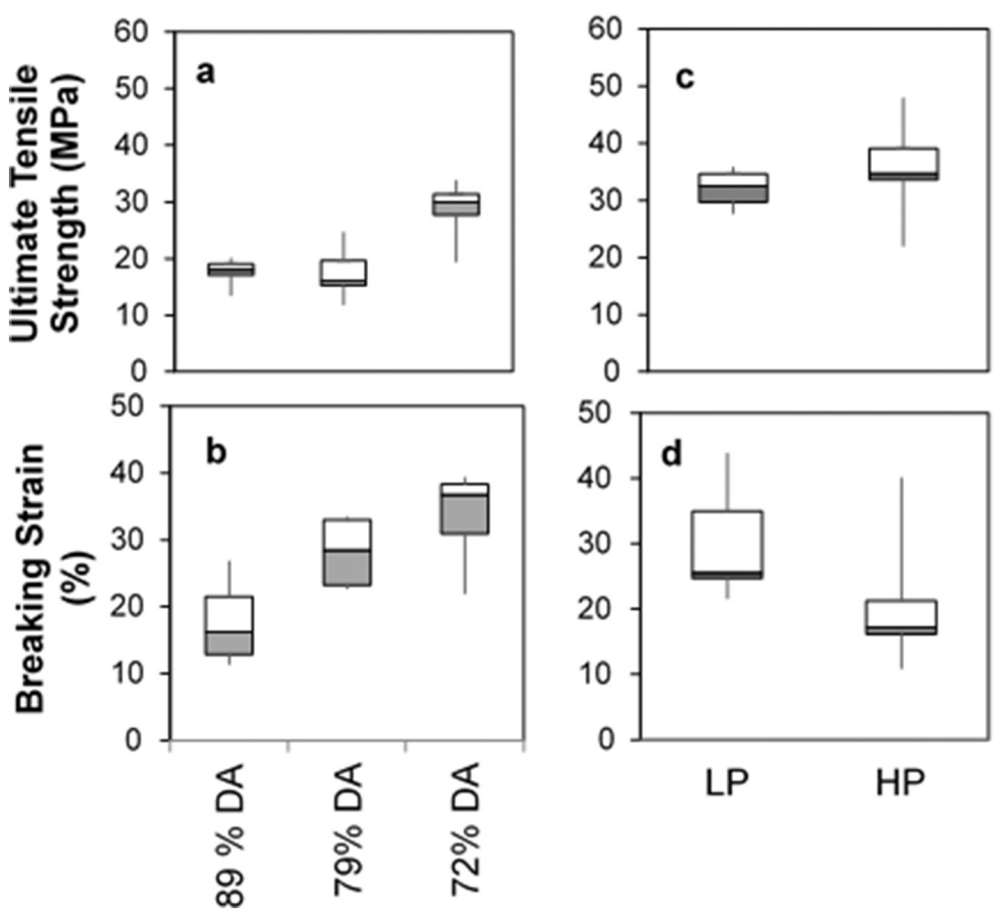

Fig. 4. (a) Ultimate tensile strength and (b) strain at break of dried films cast from homogenized commercial ChNFs with different values of DA.

it is seen that strain at break of suspensions increases with decreasing DA values, although the values of 72% and 79% DA samples were statistically indistinguishable ($p > 0.05$).

The UTS of the fibers correlates well with their particle length distributions (Table S1), as both the UTS and particle sizes of the 79% and 89% DA samples are quite similar. The dependence of tensile strengths on fiber lengths have been reported before for unidirectional plys where increase in UTS with decrease in fiber lengths is attributed to defects along fiber length (Vasiliev & Morozov, 2013). This is mainly for cases where the single fibers are stronger than the composite, and as fiber length reduces, the number of defects decrease leading to increased UTS. Fan et al. (Fan, Fukuzumi, Saito, & Isogai, 2012) reported that surface deacetylated α-chitin nanofibers/nanowhiskers had significantly higher tensile strengths compared to regenerated chitin (amorphous) due to their higher crystallinity. The ChNFs in general are known to have crystalline regions with very high tensile strength separated by amorphous domains (Oh et al., 2016). In addition it is known that the *N*-deacetylation of chitin preferentially occurs in the amorphous domains (Bradic, Bajec, Pohar, Novak, & Likozar, 2018) leading to higher *N*-deacetylation of the amorphous domains (and conversion to chitosan) making them susceptible to dissolution in acidic HPH medium. The dissolution of these "weaker" amorphous regions would explain the particle length reduction leaving behind more crystalline regions that would be relatively defect free explaining the increased UTS. As seen in Table S2, and Fig. S2(b) the HPH using deacetylated ChNFs leads to more fibrillation leading to lower fiber widths. The lower fiber widths along with higher surface charge (Table 1) may lead to better packing and lower number of large randomly aggregated fibers leading to more effective stress distribution to surrounding load bearing fibers in case of random fiber breaks and consequently higher overall strain at break of deacetylated ChNF samples.

Fig. 4(c) and (d) shows the UTS and strain at break, respectively, of films cast from suspensions produced by homogenizing chitin at different pressures. It is seen that the LP chitin samples has an average UTS value 32 MPa and is similar to HP chitin samples with average UTS value of 36 MPa. LP samples have a mean strain at break values of 30%. HP chitin samples have lower strain at break with a mean value of 20%. However, these strain at break values are not significantly different (Table S6, $p = 0.014$). The similar values are expected due to similar length and width distributions and similar alignment produced since the DA of the chitin is similar.

3.6. Influence on film surface and internal formation

Fig. 5(a–c) shows the AFM height and phase images of films cast from homogenized chitin suspensions having different DA values. The images show entangled chitin fibers as sheets. 89% DA samples show large fibers both height and phase image. The number of larger fibers is less in the 79% DA height and phase images. The 72% DA images show a few large fibers, which are seen to have a smaller width than the larger fibers seen in the 89% DA images. The roughness of the films decrease with DA and this trend corresponds well to peak fiber widths reported in Fig. S2 (b). Fig. 5(d) and (e) shows that homogenization at different pressure has no significant effect on roughness, and that the HP and LP samples themselves appear very similar, in good agreement with the similar size range of fiber lengths and widths discussed previously (Fig. S3).

Fig. 6(a–c) shows SEM cross sectional images of films with different DA, which all show lamellar structures throughout the thickness. Films with 89% DA show lamella that are loosely bound to each other, but the interlamellar distance is reduced as DA decreases, with the 72% DA films showing almost no delamination.

Decrease in interlamellar spacing (or tighter interlamellar stacking) can be explained by the decreasing fiber widths in Fig. S2(b) coupled with increase in cationization (Table 1). The decrease in spacing with increased cationization of the fibers seems runs contrary to the expectation that higher repulsion would lead to increased spacing, however it

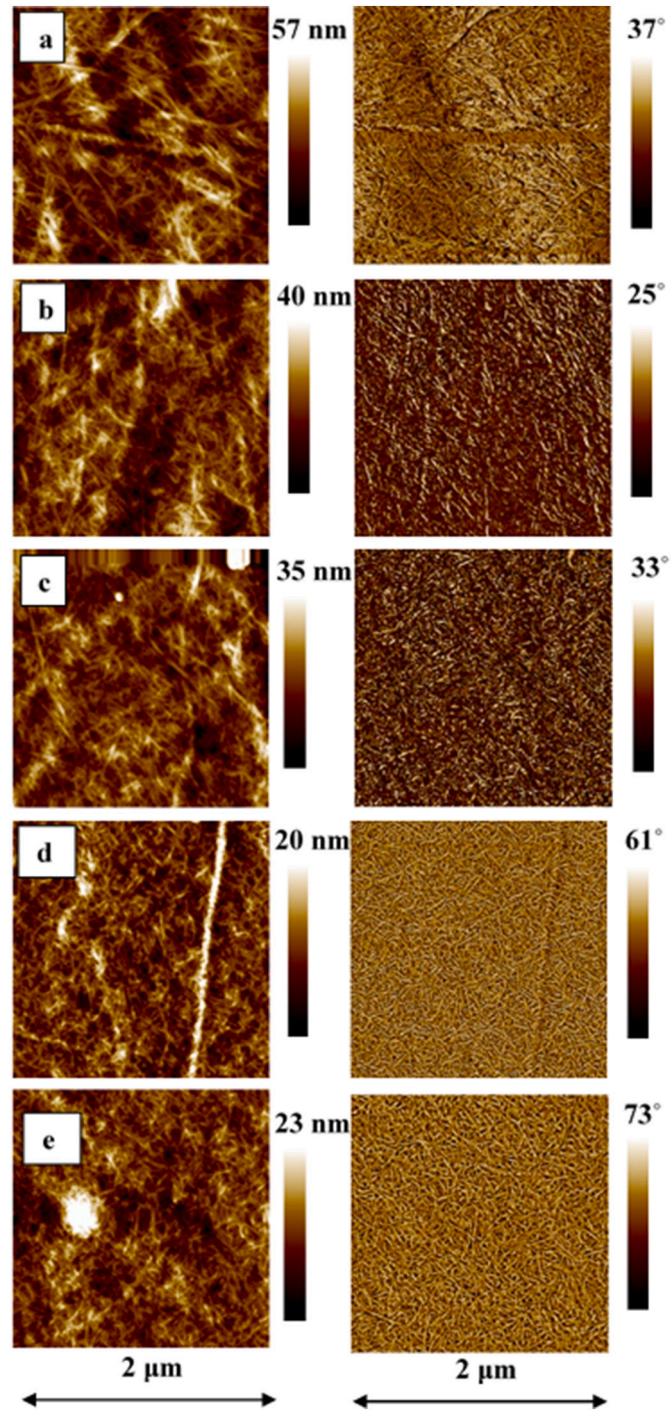

Fig. 5. AFM images of the top surface of dried cast films of (a) 89% DA, (b) 79% DA, and (c) 72% DA ChNFs processed in a two-pressure cycle as in experimental methods and of a 72% DA sample processed at constant pressure of (d) LP = 551 bar or (e) HP = 1723 bar. Height images are on the left and phase images are on the right-hand side.

can be explained as follows. In absence of the charges, the alignment of fibers in suspension is largely driven by hydrogen bonding. Fiber-fiber collisions in the suspensions lead to formation of loose aggregates largely driven by hydrogen bonding between fibers.

The charge driven repulsions help overcome the weaker hydrogen bonding in a suspension of charged fibers effectively driving the alignment. Increased cationization leads to better alignment of charged fibers in suspensions as fibers are forced into optimal configurations that

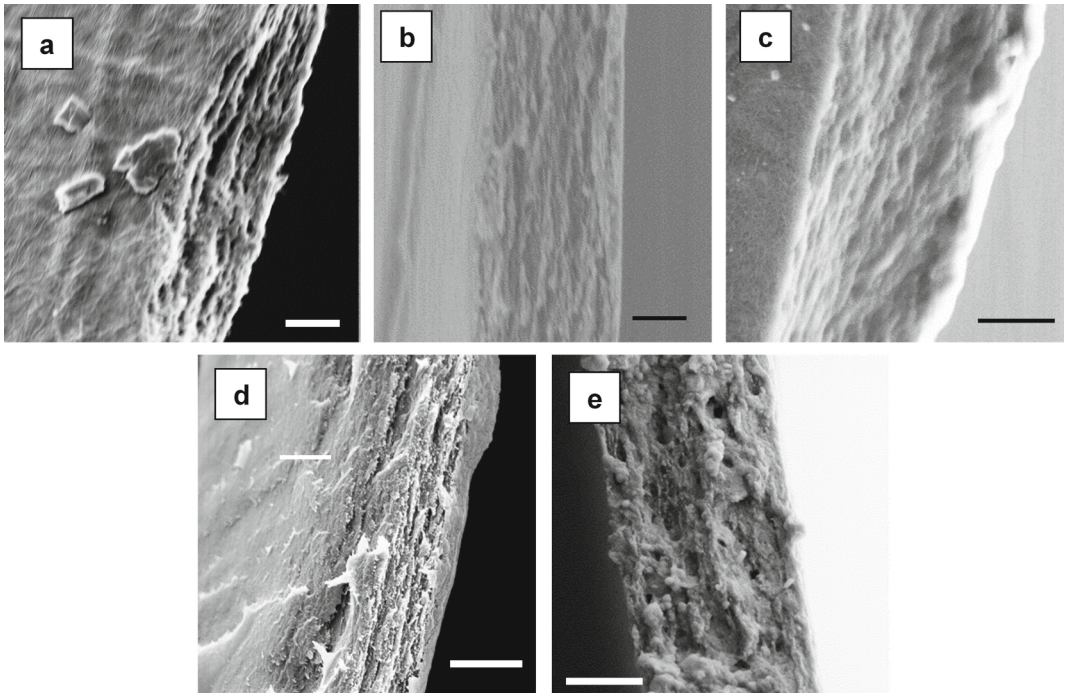

Fig. 6. SEM images of cross-sections of (a) 89% DA, (b) 79% DA, and (c) 72% DA ChNF films processed in a two-pressure cycle as in experimental methods and of a 72% DA sample processed at constant pressure of (d) LP = 551 bar or (e) HP = 1723 bar. In (b), (c) scale bars are 500 nm and in (a), (d) and (e) scale bars are 1 μm.

minimize potential energy in a field of highly charged fibers. As water in spaces between fibers evaporates during drying, the removal of water leads to the fibers being forced into close vicinity of each other. In the case of highly charged fibers, they are still more aligned than less charged fibers due to the strength of the repulsive interactions. This leads to better alignment in the dried films for highly charged fibers as compared to less charged ones.

Fig. 6(d–e) shows the SEM cross sectional images of films produced from ChNFs under different homogenization pressures. LP and HP chitin films are similar to the 72% DA chitin films seen in Fig. 6(c) due to the DA being the same. Thus, cationization of the fibers is the same which drives similar alignment.

To provide continuity with prior work that utilized ChNFs from crab shell chitin (Satam et al., 2018; Satam et al., 2019; Wu et al., 2014), crab shell chitin was deacetylated by using the procedures applied to 72% DA shrimp-derived chitin above. The resulting DA of the crab shell chitin was 64%. The ability to homogenize and achieve reduced O_2 barrier properties was assessed for the 64% DA chitin and compared to unmodified 91% DA chitin. As described in supplementary information Fig. S4, the number of passes required to achieve ChNFs with equivalent suspension light-transmission and pH was reduced from 30 (91% DA) to 8 (64% DA), even considering the 8 passes were performed at reduced pressure of 551 bar. In addition, the 8-pass homogenization of 64% DA crab shell chitin resulted in ChNF films with OP of 1.9 ± 0.7 cm^3-μm/m^2/day/kPa, compared to 2.5 ± 0.8 cm^3-μm/m^2/day/kPa for 91% DA chitin (Table S9). There was a corresponding increase in tensile strength and strain-at-break for the 64% DA, 8-pass ChNFs compared to 91% DA, 30-pass ChNFs (Fig. S5). These results for crab shell chitin mirror the effects of deacetylation on homogenization, barrier and mechanical properties observed for shrimp-derived commercial ChNFs.

4. Conclusion

The homogenization step is one of the most intensive in the production of ChNFs, due to the low solids loadings, high pressures and multiple passes required. This study shows that the efficiency of HPH can be improved when deacetylation pretreatment is used to increase cationization of ChNF fibers. Increased deacetylation intensity is found to improve the optical transmission and mechanical properties of films, and result in similar oxygen barrier properties. With the highest extent of deacetylation (72% DA), and the use of low homogenization pressure produced films with equivalent optical, mechanical and barrier properties as at high pressure, which is a significant result of the study. This is a result of similar size distributions of ChNF fibers produced at LP and HP. The OP values of the LP and HP chitin samples were 1.4 to 1.6 cm^3-μm/m^2/day/kPa, which is ~10 times lower than biaxially oriented poly (ethylene terephthalate). In addition to allowing pressure reduction, we showed that increased deacetylation also allows the number of homogenization passes to be reduced significantly (from 30 to 8 passes), even at a reduced pressure of 551 bar. These results show that intensity of HPH can be decreased significantly by deacetylation pretreatment, without affecting the already low oxygen barrier values, and producing improvements in optical and mechanical properties.

CRediT authorship contribution statement

Chinmay C. Satam: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **J. Carson Meredith:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare no competing interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.carbpol.2021.118658.

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